



KUMAR VAIBHAV SHARMA

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Summary

Driven and accomplished Data Engineer and ETL Developer with a remarkable 10+ years of hands-on experience in crafting robust data solutions. Certified GCP professional Data Engineer and AWS Solution Architect, led mission-critical projects with a proven track record of cost-saving initiatives, performance enhancements, and automation. Expertise spans the spectrum of data engineering, encompassing data migration, ETL process optimization, and building scalable data pipelines. Proficient in Python, SQL, and Spark, excel at transforming complex data into actionable insights. Committed to delivering high-quality, efficient, and secure data solutions has consistently driven organizational success.

Skills

Skills: Python, Scala, Unix, SQL, HQL, Jenkins, Terraform, GIT, GITHUB, AWS (S3, RedShift, Glue, Lambda, Athena), GCP, Jupyter, Visual Studio Code, PyCharm, HIVE, No SQL (HBASE), HADOOP, Airflow, Apache Spark, Apache Kafka, PySpark, Spark SQL, GCP DataProc, Dataflow, Informatica PowerCenter, Looker, DBT, DataForm, Azure Datafactory, Azure Datalake Storage, Azure Data Factory.

Experience

Cloud Data Platform Engineer Ford

09/2020 to Current
Dearborn, MI

- Participated in the successful migration of Teradata to BigQuery, utilizing BigQuery Data Transfer Service and Cloud Storage, resulting in an impressive annual cost reduction of \$552,000 and a remarkable 12% increase in system performance.
- Achieved a 20% reduction in monthly manual workload by implementing automated ETL processes, effectively handling billions of data rows.
- Utilized Python with Spark to distribute data processing for large streaming datasets, resulting in a substantial 56% improvement in data ingestion and processing speed.
- Implemented and Maintained Google Cloud infrastructure, including VPC, load balancers, firewalls and DNS, resulting in 25 % Deployment time and improved overall system reliability.
- Developed Terraform module for automated infrastructure provisioning and consistent deployment.
- Worked on CI/CD pipeline and Git Version to automate migrations in Tekton for Ford Credit which reduced 30% deployment errors
- Implemented CI/CD pipelines integrating Terraform and Docker with GitHub Actions, automating the build, test, and deployment workflows, resulting in faster and more efficient delivery of updates and infrastructure changes.
- Enhanced cloud platform security by implementing role-based access controls using IAM, strengthening VPC security configurations, and managing encryption keys with KMS, resulting in improved data protection and compliance adherence.
- Led initiatives to harden Linux environments by configuring and managing Pluggable Authentication Modules (PAM), securing SSH Daemon (SSHD), and setting up audit for enhanced security posture and threat mitigation.
- Spearheaded the adoption of Infrastructure as Code practices using Terraform to automate infrastructure provisioning, configuration, and management, resulting in streamlined deployment processes and increased operational efficiency.
- Applied knowledge of common network protocols to implement secure data transmission mechanisms, ensuring data confidentiality and integrity in line with industry standards and regulatory requirements.
- Demonstrated proficiency in container technologies by deploying applications on Kubernetes (k8s), optimizing resource utilization, and implementing security best practices to facilitate scalable and resilient containerized environments.
- Conducted thorough platform change reviews for infrastructure security concerns, identified potential vulnerabilities, communicated findings to application teams effectively, and provided infrastructure support to ensure system security and resilience.
- Monitor OS resources (CPU, disk, memory, and network) and perform root-cause analysis and recommendations for technical issues
- Support and troubleshoot scripts. Assist system administrators with integration and implementation of automation.

Data Engineer

02/2020 to 09/2020

Toyota Financial Services

Dallas, TX

- Achieved an impressive data pipeline up-time of 84%, efficiently handling streaming and transactional data ingestion from five different primary data sources using combination of Spark, Python, and S3 for seamless data management.
- Spearheaded the creation of data pipelines, leveraging SQL and Spark, to establish a robust Big Data ecosystem that integrated Python, Hadoop, Spark, NoSQL, and various other essential tools.
- Developed a robust CI/CD pipeline on Jenkins, streamlining the code migration process to higher environments, enhancing software deployment efficiency.
- Proficiently crafted SQL queries to provide valuable business insights through data analysis.

Data Engineer/ETL Developer

02/2019 to 01/2020

TD Ameritrade

Ellicott City, Maryland

- Spearheaded the design and development of a robust ETL (Extract, Transform, Load) strategy, enabling efficient data population from diverse source systems.
- Conducted thorough data analysis across multiple source systems, paving the way for precise source-to-target mapping creation to facilitate data warehouse loading.
- Leveraged Informatica 10 extensively for data loading, seamlessly handling various data sources, including flat files and Oracle databases.
- Proactively monitored ETL production batch jobs, investigating, and resolving job failures promptly to maintain data pipeline reliability and performance.

Data Engineer/ETL Developer

03/2018 to 01/2019

CareFirst BCBS

Baltimore, Maryland

- Designed and implemented ETL strategy using Informatica 10 for diverse source systems

- Developed complex data loading processes for ODS, Data Warehouse, and Data Marts.
- Utilized Informatica to create mappings for reading and generating XML files.
- Optimized performance of mappings, transformations, and workflows.

ETL Developer

11/2016 to 03/2018

Prescience Corporation

Dallas, TX

- Gathered business requirements, managed junior developers, and crafted ETL mappings with Informatica for SAP data extraction and transformation.
- Automated ETL processes, maintained data warehouse standards, and conducted data validation using HP Quality Center, Unix scripts, and IDQ data profiling.

ETL Developer

04/2016 to 10/2016

Cymansys Solutions

Austin, TX

Education

Master of Science: Computer Science (December 2015)

Governor's State University

University Park, IL

Bachelor of Technology: Information Technology

JNTU

Hyderabad, India

Certifications

- GCP Professional Data Engineer | AWS Solution Architect certificate